

User Table

All roles share a single base table. This stores authentication credentials, the role assignment, and account status.

Field	Type	Constraints	Description
user_id	UUID / String	PRIMARY KEY	Unique system identifier for each user
name	String	NOT NULL	Full legal name of the user
email	String	UNIQUE, NOT NULL	Used as the login identifier
password_hash	String	NOT NULL	Bcrypt-hashed — never stored as plain text
role	ENUM	NOT NULL	One of: LECTURER HOD ADMIN
is_active	Boolean	DEFAULT TRUE	Soft-disable accounts without deleting history
created_at	Timestamp	NOT NULL	Date and time the account was created
updated_at	Timestamp	NOT NULL	Date and time of last modification

Role-Specific Profile Tables (Separate Table Approach)

In the separate table approach, the base User entity is extended with one dedicated profile table per role. Each profile table holds a one-to-one relationship with User by user_id.

LecturerProfile

Stores academic and contact details specific to lecturers.

Field	Type	Constraints	Description
lecturer_profile_id	UUID	PRIMARY KEY	Profile-level identifier
user_id	UUID	FK -> User, UNIQUE	Links to the base User record (1:1)
staff_id	String	NOT NULL	Institutional ID e.g. UTM/LEC/0042
department	String	NOT NULL	e.g. Computer Science
phone_number	String	NULLABLE	Contact number for administrative use
joined_date	Date	NULLABLE	Date the lecturer joined the institution

HODProfile

Stores program oversight information for the Head of Department.

Field	Type	Constraints	Description
hod_profile_id	UUID	PRIMARY KEY	Profile-level identifier
user_id	UUID	FK -> User, UNIQUE	Links to the base User record (1:1)
staff_id	String	NOT NULL	Institutional ID e.g. UTM/HOD/0005
program_name	String	NOT NULL	e.g. Diploma in Computer Science
department	String	NOT NULL	The department being overseen
office_location	String	NULLABLE	Optional room or block reference

AdminProfile

Stores system administration scope and access level details.

Field	Type	Constraints	Description
admin_profile_id	UUID	PRIMARY KEY	Profile-level identifier
user_id	UUID	FK -> User, UNIQUE	Links to the base User record (1:1)
staff_id	String	NOT NULL	Institutional ID e.g. UTM/ADM/0003
access_level	ENUM	NOT NULL	SUPER_ADMIN ADMIN
managed_since	Date	NULLABLE	Date admin rights were granted